

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

DECEMBER/JANUARY EXAMINATION

BMIT3084 ENTERPRISE NETWORKING

WEDNESDAY, 7 JANUARY 2026

TIME: 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN INFORMATION SECURITY
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BMIT3084 ENTERPRISE NETWORKING**Question 1**

As a network engineer, your task is to analyse, illustrate and suggest different types of static routes configurations that can be applied for a small network as shown in Figure 1-1 to your team members and management. All devices in Figure 1-1 network topology are configured with Internet Protocol version 4 (IPv4) addressing.

Assume a **fully-specified default static route** is pre-configured in **MLS** and also a **default static route** is pre-configured in **Healthcare_RTR**. A default static route is also pre-configured in **Wellness_RTR** to forward traffic to **Medical_RTR**. In addition, **standard static routes and floating standard static routes** were pre-configured in the **ISP** to forward packets to **Healthcare_Dept** and **Wellness_Dept**.

Refer to Figure 1-1 and Figure 1-2 to answer the following questions.

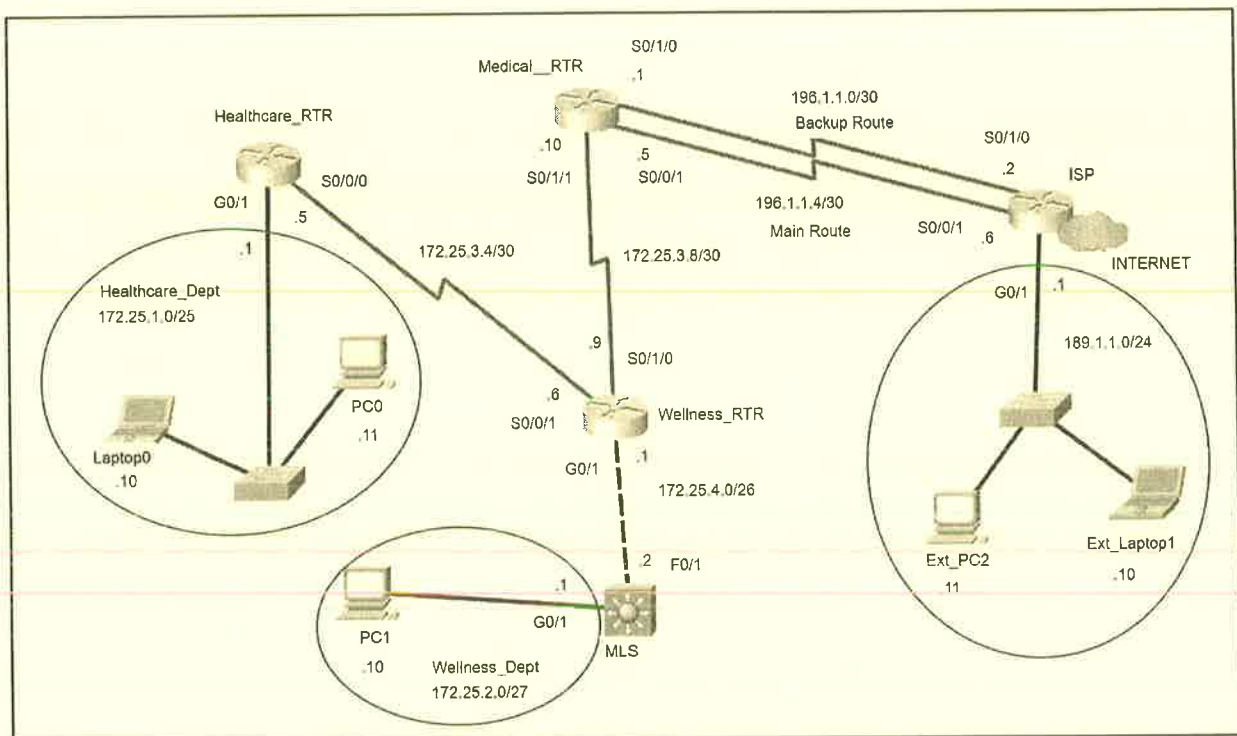


Figure 1-1: A network topology

```
Medical_RTR#show ip route
```

```
Gateway of last resort is 196.1.1.2 to network 0.0.0.0
```

```
172.25.0.0/16 is variably subnetted, 4 subnets, 4 masks
```

```
C 172.25.3.8/30 is directly connected, Serial0/1/1
```

```
196.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C 196.1.1.0/30 is directly connected, Serial0/1/0
```

```
S* 0.0.0.0/0 [22/0] via 196.1.1.2
```

```
:
```

Figure 1-2: Partial output for show ip route

BMIT3084 ENTERPRISE NETWORKING**Question 1 (Continued)**

- a) (i) Configure a **default static route** and a **floating default static route** using next hop IP address in **Medical_RTR** to forward packets to the **ISP** and the **INTERNET**. Refer to Figure 1-2 for the AD value. (4 marks)
- (ii) Give **THREE (3)** explanations for the output of **S* 0.0.0.0/0 [22/0] via 196.1.1.2** as shown in Figure 1-2. (6 marks)
- (iii) Configure a **fully specified standard static route in Wellness_RTR** to forward traffic to **Wellness_Dept**. (4 marks)
- (iv) Configure a **standard static route in Wellness_RTR** to forward traffic to **Healthcare_Dept**. (3 marks)
- b) Your company management is planning to expand its business in future and static routing configurations are not feasible for a large organisation. A dynamic routing protocol is more suitable for larger organisation. Assume that only a **default static route** and a **floating default static route** are configured on **Medical_RTR** to forward packets to the **ISP** and the **INTERNET**, and the **ISP's** static routing pre-configuration remains unchanged. The static routing configurations in **Healthcare_RTR**, **Wellness_RTR**, **Medical_RTR** and **MLS** are removed in Figure 1-3 network topology. Assume the "ip routing" command is enabled in **MLS**.

Now, you are required to apply a dynamic routing protocol called **Open Shortest Path First (OSPF)** using the network command with a wildcard mask based on the subnet mask in **Medical_RTR** and **MLS** only. Propagate the default static routes in **Medical_RTR** to **Wellness_RTR**, **MLS**, and **Healthcare_RTR** to forward traffic to the **ISP** and the **INTERNET**. Use **OSPF process-id 1357** and **area-id 0**. Use Table 1-1 to document your configurations.

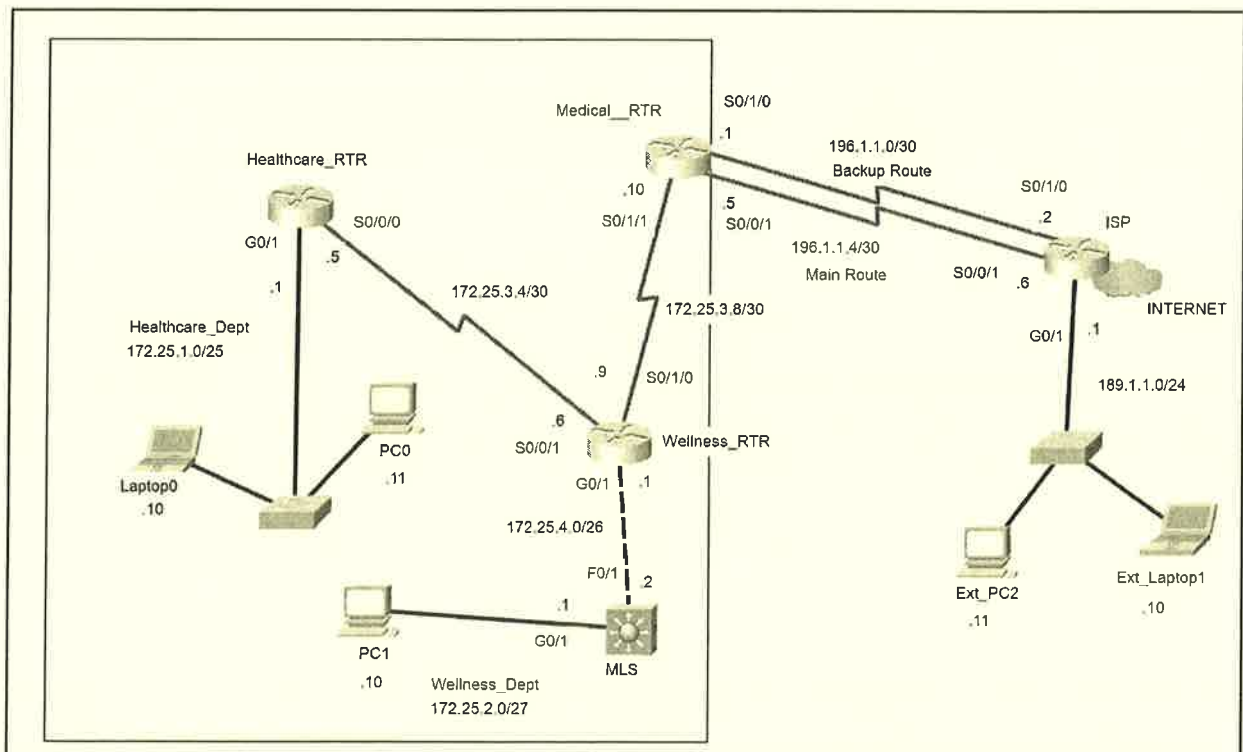


Figure 1-3: A network topology

BMIT3084 ENTERPRISE NETWORKING**Question 1 b) (Continued)**

Table 1-1: Documentation Table

Router name	Configurations

(8 marks)

[Total: 25 marks]

Question 2

- a) Illustrate how an ARP cache poisoning attack occurs by including the diagram from Figure 2-1 in your answer. Also include the ARP cache tables for PC-A and R1 after the attack.

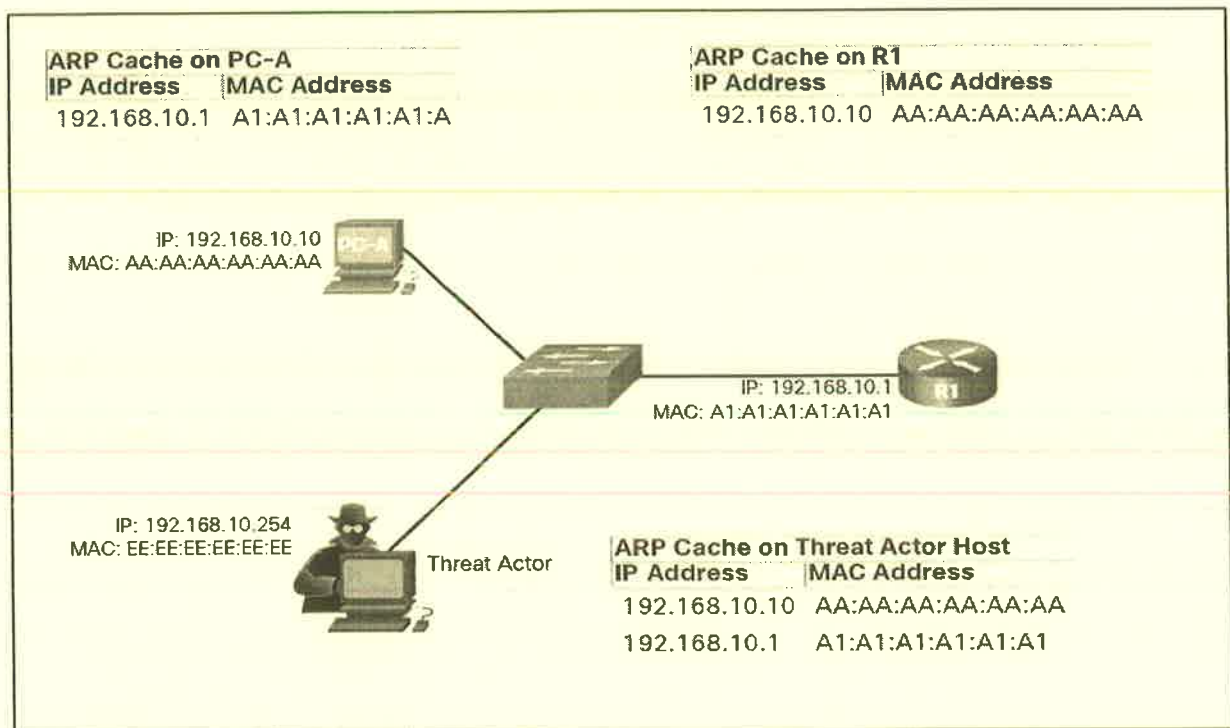


Figure 2-1: ARP cache poisoning attack

Reference: Cisco Networking Academy. (2025). *Module 3: Network security concepts – 3.8.2 ARP cache poisoning*. <https://www.netacad.com>

(10 marks)

BMIT3084 ENTERPRISE NETWORKING**Question 2 (Continued)**

- b) Figure 2-2 shows a network topology in which all hosts, routers, and the Layer 3 switch are able to communicate with one another. Examine the network topology to apply Access Control Lists (ACLs) in the following question.

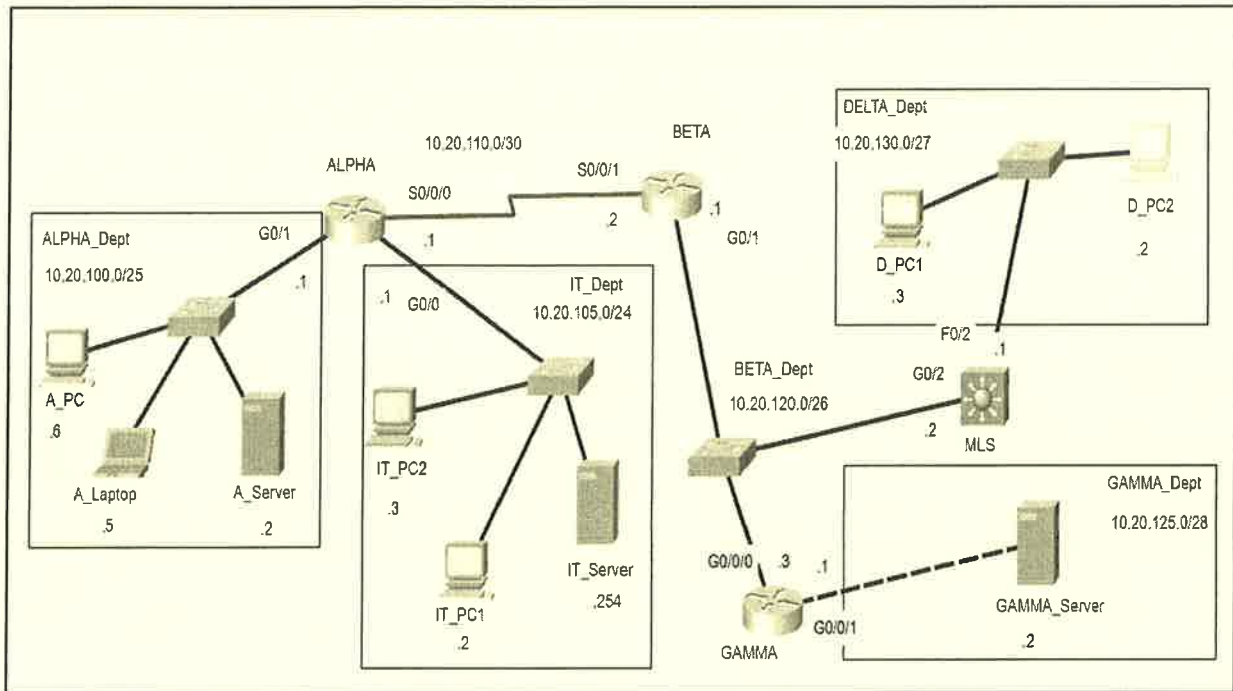


Figure 2-2: A network topology

- (i) Write an extended access list numbered 176 to allow odd numbered IP addresses from the **IT_Dept** to access the **GAMMA_Server** via **HTTPS**. Deny all other traffic, which must be explicitly written in your ACL. Use keywords in your ACL, except for port number 443 for **HTTPS**. Indicate the **router**, **interface** and **direction** to apply the ACL efficiently. (10 marks)
- (ii) Describe **TWO (2)** differences of applying an access list on incoming or outgoing port of a router. (5 marks)

[Total: 25 marks]

BMIT3084 ENTERPRISE NETWORKING**Question 3**

In Figure 3-1 network topology, analyse and apply configurations for Dynamic Host Configuration Protocol (DHCP), Static Network Address Translation (NAT) and Dynamic Network Address Translation (NAT). Assume OSPF and static routing protocols are configured in the respective routers and Layer 3 switch.

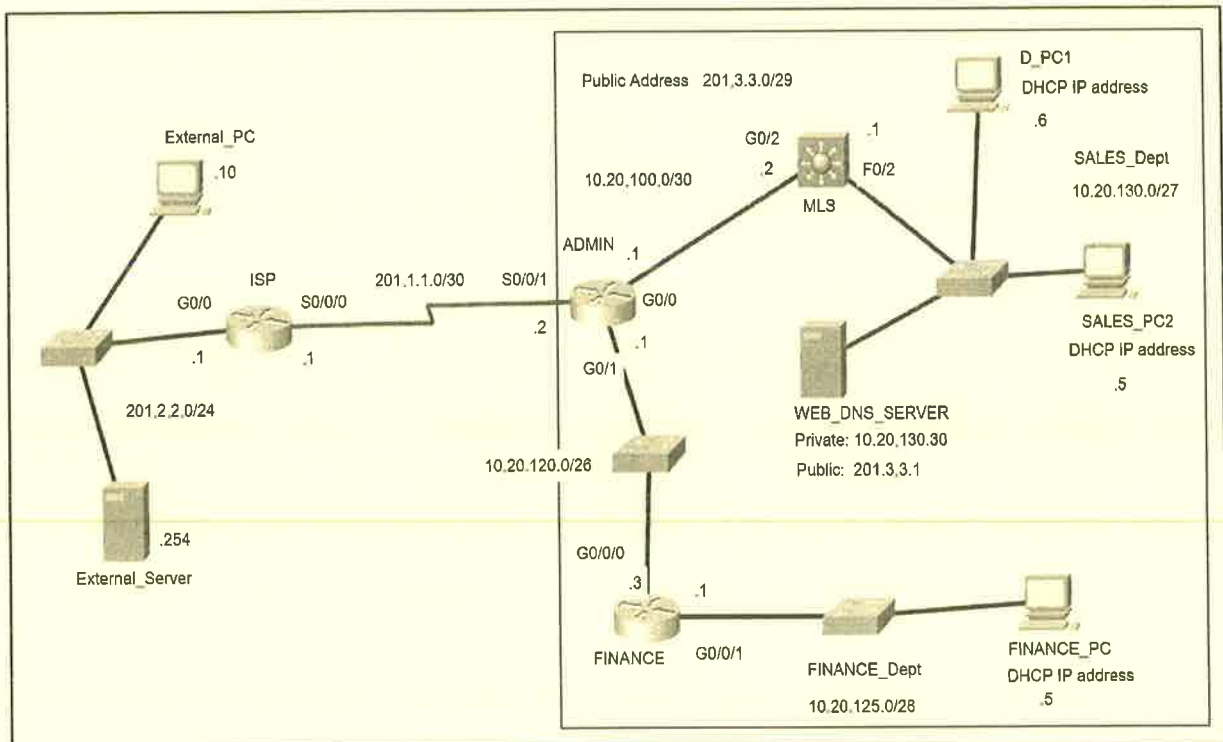


Figure 3-1: A network topology

FINANCE	MLS
<pre> ip dhcp excluded-address 10.20.125.1 10.20.125.4 ip dhcp excluded-address 10.20.130.1 10.20.130.4 ip dhcp pool FINANCE_POOL network 10.20.120.0 255.255.255.192 dns-server 10.20.130.30 ip dhcp pool SALES_POOL network 10.20.130.0 255.255.255.224 dns-server 10.20.130.30 interface GigabitEthernet0/0/0 ip address 10.20.120.3 255.255.255.192 interface GigabitEthernet0/0/1 ip address 10.20.125.1 255.255.255.240 </pre>	<pre> interface FastEthernet0/2 no switchport ip address 10.20.130.1 255.255.255.224 interface GigabitEthernet0/2 no switchport ip address 10.20.100.2 255.255.255.252 </pre>

BMIT3084 ENTERPRISE NETWORKING**Question 3 (Continued)**

ADMIN
interface GigabitEthernet0/0 ip address 10.20.100.1 255.255.255.252
interface GigabitEthernet0/1 ip address 10.20.120.1 255.255.255.192
interface Serial0/0/1 ip address 201.1.1.2 255.255.255.252 clock rate 2000000
ip nat pool NAT_POOL 201.3.3.2 201.3.3.7 netmask 255.255.255.240
access-list 32 permit 10.20.130.0 0.0.0.31

Figure 3-2: Partial output of “show run” command

- a) (i) Examine the partial output of the “show run” command in Figure 3-2 and also the network topology in Figure 3-1. The **FINANCE** router is configured as a DHCP server. **FINANCE_PC**, **D_PC1** and **SALES_PC2** are having problems obtaining the IP addresses and other DHCP configurations successfully. Use Table 3-1 to document all errors, provide the solutions or correct configurations for the respective errors and justify your answers. State your assumptions in your answers, if any. (6 marks)

Table 3-1: Documentation Table

Errors	Solutions

- (ii) Explain DHCPv4 relay agent with reference to Figure 3-1 and Figure 3-2. (4 marks)
- b) (i) An organisation has been assigned a public address of **201.3.3.0/29**. The first usable public address of 201.3.3.1 is assigned to the **WEB_DNS_SERVER**. Implement **static NAT** configurations on the appropriate router and apply them to all the router’s interfaces for the **WEB_DNS_SERVER** to be reachable from the Internet. (5 marks)
- (ii) Assuming the static NAT configurations in Question 3 b) (i) have been successfully implemented. Now, you are required to identify errors and provide solutions for **Dynamic NAT** configurations using the **remaining** public IP addresses from the external network address **201.3.3.0/29** as the address pool. Use standard numbered access-list **32**. All internal PCs should be able to ping the **External_PC**. Use Table 3-2 to document your answers.

Table 3-2: Documentation Table

Errors	Solutions

(10 marks)

[Total: 25 marks]

BMIT3084 ENTERPRISE NETWORKING**Question 4**

- a) Based on Figure 4-1 and Figure 4-2, identify and explain the errors and provide solutions for VPN configurations by using Table 4-1.

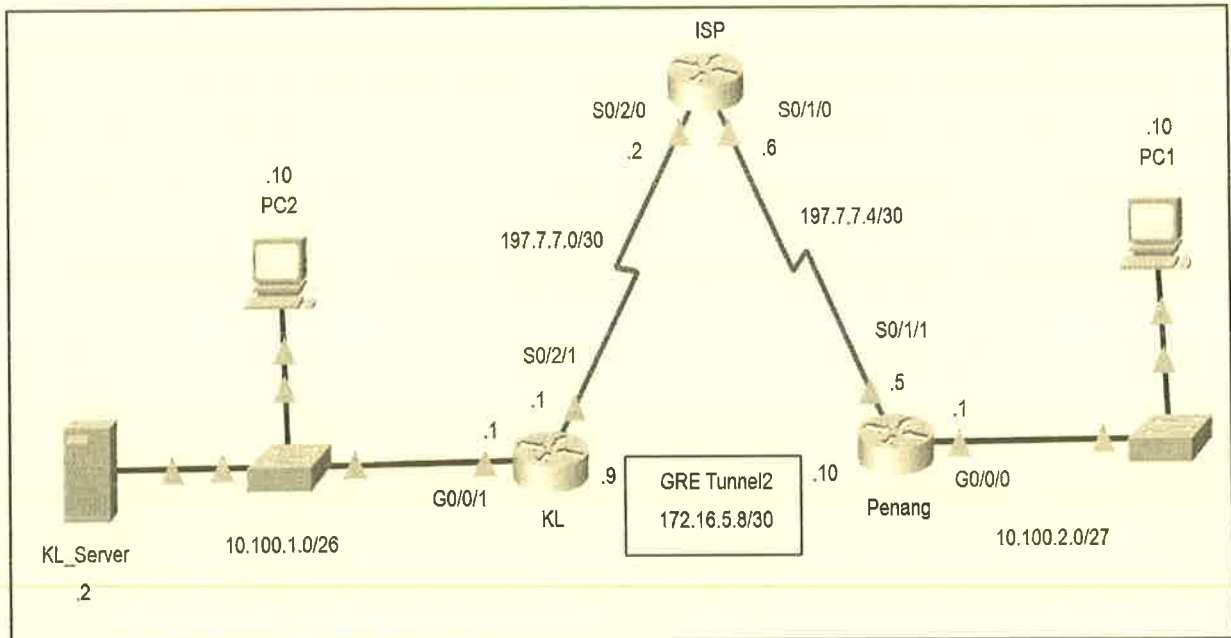


Figure 4-1: A network topology

KL	Penang
interface Tunnel2 ip address 172.16.5.8 255.255.255.252 tunnel source Serial0/2/0 tunnel destination 10.100.2.1	interface Tunnel2 ip address 172.16.5.10 255.255.255.252 tunnel source Serial0/1/0 tunnel destination 10.100.1.1
interface GigabitEthernet0/0/1 ip address 10.100.1.1 255.255.255.192	interface GigabitEthernet0/0/0 ip address 10.100.2.1 255.255.255.224
interface Serial0/2/1 ip address 197.7.7.1 255.255.255.252 clock rate 2000000	interface Serial0/1/1 ip address 197.7.7.5 255.255.255.252
ip route 10.100.2.0 255.255.255.224 197.7.7.2	ip route 10.100.1.0 255.255.255.192 197.7.7.6
ip route 0.0.0.0 0.0.0.0 197.7.7.2	ip route 0.0.0.0 0.0.0.0 197.7.7.6

Figure 4-2: Partial VPN configurations for KL and Penang

Table 4-1: Documentation Table

Router Name	Errors	Solutions

(10 marks)

BMIT3084 ENTERPRISE NETWORKING**Question 4 (Continued)**

- b) A company is planning to use Multiprotocol Label Switching (MPLS) to connect branches across different cities. Illustrate MPLS concepts and explain why it is used in modern Wide Area Network (WAN) environments. (6 marks)
- c) Your company network is experiencing voice call dropouts during peak hours. Explain which Quality of Service (QoS) techniques you would apply to improve voice quality and why. (9 marks)

[Total: 25 marks]